Principles of communication systems

EET3202, CUNY City Tech, Fall 2023

Homework #08 (Due on Nov 16)

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Problem 1

The analog message signal m(t) is digitized using a Delta modulator.

$$m(t) = A_1 \cos(\omega_1 t) + A_2 \cos(\omega_2 t)$$

$A_1 = 3$, $A_2 = 2$, $\omega_1 = 2\pi \times 10^3$, $\omega_2 = 2\pi \times 2 \times 10^3$

1. Calculate the maximum rate of change (time derivative of m(t))

2. Calculate minimum size of quantization level (E) for a sampling rate $f_s = 8$ kHz.

* hint; see lecture #10 notes.

Problem 2

Draw the different line codes discussed in class for the binary sequence 0101110010.

Line codes: On-off (RZ), Polar (RZ), Bipolar (RZ), On-off (NRZ), polar (NRZ), and the Manchester code.

Hint: see lecture #11 notes.